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Workpackage D

Smart Energy

Deliverable 1

Implementation of smart meter data in the production of official statistics (actions 1-5)

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ABSTRACT

The Smart Energy workpackage (WPD) is one of the workpackages in the ESSnet Big Data II project which aims to implement smart meter data for production of official statistics. This report is the first report out of three. The report gives a brief overview of the statistical products that will be produced by WPD, also a modified statistical production process (GSBPM) is proposed which is adapted for smart meter data. The report further covers micro and macro quality indicators that refer to the input data quality and can be used as a general quality checklist for smart meter data. The report also includes information about metadata and additional information that can be linked from other administrative sources. The report ends with a risk plan where the main risks that can affect WPD are described.

INTRODUCTION

The use of smart electricity meter data and appropriate analytical methods will enable the ESS to produce new kinds of statistics - trusted smart statistics.

A smart electricity meter measures electricity consumption at a high frequency and communicates the information to a central system. The aim for work package D is to implement smart meter data for producing statistics in different areas, e.g. energy statistics of businesses, census statistics, household costs, tourism seasonality, or impact on environment. The implementation will include linking electricity data with other administrative sources for producing statistics of businesses and households, and identifying vacant living places or seasonal/temporary occupancy of living places. The duration of the project is 24 months and there are four participating countries (Estonia, Denmark, Norway and Sweden). The work is

based on the results of the previous ESSnet and results described in previous reports [1;2;3;4] will be used as a starting point to develop functional production prototypes.

It is a great advantage for statistics production if smart meter data are available through a central national hub. Currently, Denmark, Estonia and Norway have national hubs in operation, while Sweden will have a national hub ready by 2021.

The aim of the WP is to develop functional production prototypes from smart meter data. During the project the implementation procedures for the following statistical products will be produced:

1. electricity statistics of businesses, by sector (EE, DK, SE)
2. electricity statistics of households (DK, SE)
3. identifying vacant or seasonally vacant dwellings by new estimations models (EE, NO).

STATISTICAL PRODUCTION PROCESSES (GSBPM)

To describe the statistical production processes of smart meter data in a general and process-oriented way the Generic Statistical Business Process Model (GSBPM) will be used. GSBPM is the standard adopted for describing production activities performed by NSIs. The principal phases of GSBPM are shown in Figure 1.

Figure 1: Principal phases of GSBPM (Istat 2018)

In order to use GSBPM model in a smart energy context, a small model adaptation is needed. The Italian Statistics Agency ISTAT suggests in their Big Data Committee report to swap the Design and Collect phases, because only after collection of the data sources, the necessary framework for data processing can be set up. The Design phase can be considered as a Preparation phase, where, rather than doing traditional design choices (e.g. sampling design), a set of preparation activities have to be carried out on the Big Data source [5]. The Italian report is mainly based on experiences with web scraping activities, but the modified production process is of general relevance. The smart energy data are generated for different purposes than official statistical analysis, and they too need to be "prepared". Figure 2 shows the proposed GSBPM that is adapted to Big Data processing and in particular smart energy data processing. The changes mentioned above are highlighted in red.

Figure 2: GSBPM adapted to Big Data processing (Istat 2018)

The Preparation phase for smart energy data processing consists of additional steps, for example

- geocoding of the data
- anonymization of the data

Detailed descriptions of smart energy data processing during the preparation phase are presented in a previous ESSnet Big Data report [1].

In the GSBPM Processing phase the data are checked, cleaned, and transformed so that it can be used for statistical output production. The electricity data are linked and enriched with other administrative data sources that make them suitable for producing statistics of businesses, households and identifying vacant living places.

During the Analyze phase, statistical outputs are produced and examined in detail. Different classification methods are used to identify characteristics of the smart meter data. It includes the mapping between an observed unit - a metering point - and a statistical unit - a business, household or dwelling and assuring good quality outputs from the linked data. The identification of the end consumer can be problematic because consumption measured at a metering point is assigned to the owner of the contract with an energy company, who is not always the actual consumer. The geocodes of the metering points and exact addresses are important in order to solve this problem. Unsupervised classification algorithms could be used to identify different types of electricity consumption patterns and analyze the possibility to discriminate businesses from private consumers if this information is missing in the smart meter data.

In the Disseminate phase the statistical products are released to the customers. For visualization, clustering and machine learning, tools like Anaconda or Spark can be used.

The evaluation phase manages the evaluation of the statistical business process. It takes place at the end of the process, but relies on inputs gathered throughout the different phases. It includes evaluating the success of a specific instance of the statistical business process, drawing on a range of quantitative and qualitative inputs, and identifying and prioritizing potential improvements.

QUALITY FRAMEWORK

A quality framework should ensure the quality of both the processes for creating the statistics, and the resulting output statistics. In the previous ESSnet Big Data, a first list of quality indicators were tested [1;2]. These indicators refer to the quality of the input data and are based on the quality performance indicators developed in the ESSnet on quality of multisource statistics [6]. The purpose is twofold; the indicators can be used for comparisons between countries, and as general quality checklist.

Micro indicators proposed by the ESSnet work package 3 are for example

- Undercoverage
    - Proportion of households and companies that do not have smart meters
    - Proportion of consumption that is not covered
- Percent of units that fail basic checks, such as checks for extreme readings
- Percent of units that are adjusted
- Percent of units imputed
- Periodicity, i e frequency of data delivery
    - What is the time between the reference time and the data delivery time. If there are more than one reference time for each delivery (e.g. if there are data for each day but there is delivery once a month)
    - What is the maximum and minimum time between the reference time and delivery time
- Delays in delivery of data
    - Delay in delivery compared to scheduled delivery: how many days after schedule are the dataset delivered
    - Percentage of data not updated at delivery

- Percent of common units, i e possible overlap if multiple sources, or duplicates from a single source
- Differences between administrative and statistical units
- The number of observations per month and per metering point
- Number of days per year and per metering point
- Number of outliers per metering point

Additional macro indicators are

- Comparison of consumption data and the annual consumption of electricity per sector as observed from the survey
- Comparison of consumption data and the output from the household energy consumption survey

At the time, Estonia and Denmark were the only countries with enough hub data available to test the indicators. Attempts to develop a quality framework were also discussed in ESSnet Big Data work package 1 but did not result in any suggested indicators [7]. Two different existing quality frameworks were investigated, the Quality Assessment Framework by Statistics New Zealand (Reid et al 2017) [8] and the Framework for the Quality of Big Data by the UNECE Big Data quality task team (UNECE 2014) [9].

The NZ framework builds on the Total Survey Error (TSE) paradigm (Biemer 2010) [10] and an extension of TSE to administrative and integrated data (Zhang 2012) [11]. It combines the classical statistical thinking of sources of error (i e accuracy) with the notion that modern statistical production involves mixing different data sources. It is not particularly tailored to Big Data sources.

The UNECE framework uses a structure with three hyper dimensions; the source, the metadata, and the data, that apply to the three phases of the business process; input, throughput, and output. The source relates to the type of data and characteristics of the objects where data are measured, as well as administration and regulations for obtaining data. The metadata refers to descriptions of concepts and content of data sets, and the hyperdimension data relates to the quality of the data itself.

Quality dimensions are nested within the hyperdimensions. Institutional environment, privacy and security are examples of quality dimensions within the source dimension. Traditional quality dimensions such as accuracy and consistency pertains to the data dimension, as does the quality of the linking variables. The metadata dimension describes for example complexity, completeness, usability, linking variables, standardisation, and transparency of methods and processes.

The UNECE framework addresses more quality type issues, compared to the NZ framework. It was concluded in work package 1 that the UNECE framework was more intuitive (for on-line job dvertisements) but that the accuracy and selectivity dimensions of the UNECE framework would benefit from the TSE approach of the NZ framework.

The TSE approach has been incorporated in a big data framework by the AAPOR Task Force on Big Data. They introduce the BDTE (Big Data Total Error) framework (AAPOR 2015) [12]. With the BDTE approach, more focus is on errors in the data generating process, compared to the TSE approach.

In their Big Data Committee Annual report for 2017, ISTAT (2018) presents thoughts on a methodology framework for Big Data. They discuss approaches to handle bias introduced by selectivity and validity concerns associated with Big Data sources.

In the present project, we will continue to work on a quality framework for electricity hub data covering not only the input data but also the process of producing statistics, and the statistical output. The work will take into account the results from WP K Methodology and quality.

METADATA

Metadata contains information available through the hub, as well as additional information linked from other administrative sources.

Hub data contains information on metering points, customers, agreements and the consumption/production of energy. This could for example be

- Registered user information: registered user id, street name, street number, house number, postal code, postal town/place
- Registered user invoice information: similar to registered user information
- Metering point information, similar to registered user information including metering point id, cadastral unit number, property unit number and the location coordinates of the metering point
- Industry code for local units/enterprises
- For other metering points: whether metering point is used for household, cabin, street light, green house etc.

Linking the original data with other data sources gives additional knowledge and enables to produce more varied outputs. In all the participating countries, administrative datasets - business register, population register, and/or register of buildings or dwellings - are used for linking. One important reason for linking to business information is to capture the NACE

(Nomenclature of Economic Activities) code for local units from where it is possible to get the consumption by economic activity. The linking can be done either by business code or by the location coordinates of metering points. From the population register, people living at the same address can be identified as a household, and the data can be linked with electricity data by address or a unique dwelling identification.

The resulting data set including the linked information can be viewed as a statistical register, available within the organisation for use in production of all kinds of statistics, i e energy statistics as well as economic indicators or environmental statistics.

RISK PLAN AND MITIGATION SCENARIOS

The main risks that can affect the delivery of results for all participating countries in workpackage D can be split into two types of risks:

Data handling risks:

- Protection of personal data fails according to GDPR and other relevant regulations. Electricity consumption data are by many users considered as private, and the perceived privacy increases if the granularity is finer. As an example, electricity consumption may reveal which periods of time that a dwelling is not used. Also, although the inference will be very uncertain, the consumer might be worried about the following: the consumption may suggest some patterns concerning time and frequency of cooking dinner, showering etc. The NSI has access to many administrative registers and have a security policy to ensure compliance with GDPR, and the risk of related to electricity data should not be different than the well-known risks. However, the consequences if

the protection should fail, could be much larger due to the perceived risk level among consumers.

- Too many steps the between primary data supplier and utilization of data could make it difficult to fully understand the data
- Linking data to other sources fails or is not of good enough quality
- Changes occur in the data structure
- Poor quality of the outputs

Administrative risks:

- Clear contract formulation and supplier contact maintenance fail. Expectations are not met
- Changes in legal aspects opening for the data provider to refuse delivering the data
- Lack of resources and people
- Technological or administrative changes that make previous work obsolete

To meet and mitigate the risks, it is important for the participating countries to set up contracts with the data providers and have good cooperation with them. We must also be able to adapt swiftly to changes in data structure and techniques, and for example investigate the use of models for estimating outputs. Also, we need to make sure that people are trained and that knowledge is transferred, to avoid losing momentum in case of changes in staff.

There are also a number of country specific risks. Norway is expecting to get data of finer granularity, if this is not possible, production might not give the expected results. Denmark had promised to deliver data to researchers, but if there is no benefit for the data provider, they might refuse delivering data. Sweden does not have a hub yet and there is a risk of major changes that will affect the possibility to use the data in the way currently intended. The country specific risks will not be handled by the project.

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